Yujie Liu (she/her)

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Research Interests

Stochastic modeling, queueing systems, and online learning, with a focus on understanding and improving decision-making under uncertainty in operational systems.

Background

Postdoctoral Research Fellow Dec 2024 – Present

- Industrial Systems Engineering and Management, National University of Singapore
- Advisors: Yunbei Xu and Vincent Y. F. Tan

Postdoctoral Research Fellow Dec 2023 – Dec 2024

- Industrial Systems Engineering and Management, National University of Singapore
- Advisor: Shuangchi He

Ph.D. in Operations Research Sep 2018 – Jul 2023

- Academy of Mathematics and Systems Science, Chinese Academy of Sciences
- Advisors: Dacheng Yao and Hanqin Zhang

B.S. in Mathematics and Applied Mathematics Sep 2014 – Jul 2018

- Jilin University, China
- Ranked 1st in the regular track of Mathematics and Applied Mathematics

Publications

1. **Yujie Liu**, Vincent Y. F. Tan, and Yunbei Xu. Sampling allocation of LinUCB: Optimal design limits in the small-gap regime. Accepted to *ACM SIGMETRICS 2027*.

2. **Yujie Liu**, Vincent Y. F. Tan, and Yunbei Xu. Finite-time minimax bounds and an optimal Lyapunov policy in queueing control. Forthcoming in *Operations Research*.
   Short version presented at the NeurIPS 2025 Workshop on ML×OR, San Diego, CA, Dec 2025.

3. **Yujie Liu**, Dacheng Yao, and Hanqin Zhang. A non-convex optimization approach of searching algebraic degree phase-type representations for general phase-type distributions. *Advances in Applied Probability*, 57(3): 1101–1137, 2025.

4. **Yujie Liu**, Dacheng Yao, and Hanqin Zhang. On substochastic inverse eigenvalue problems with the corresponding eigenvector constraints. *SIAM Journal on Matrix Analysis and Applications*, 45(3): 1689–1719, 2024.

5. **Yujie Liu**, Dacheng Yao, and Hanqin Zhang. Minimal positive realizations for linear systems with two-different-pole transfer functions. *European Journal of Control*, 69: 100720, 2023.

6. **Yujie Liu**, Minwen Niu, Dacheng Yao, and Hanqin Zhang. Stationary distributions and ergodicity of reflection-type Markov chains. *Statistics & Probability Letters*, 189: 109597, 2022.

Work in Progress

- **Yujie Liu**, Vincent Y. F. Tan, and Yunbei Xu. How LinUCB allocates samples: A fluid-limit analysis under local gaps. Target: *Operations Research*.
- **Yujie Liu**, Dacheng Yao, and Hanqin Zhang. Queue regulation based on waiting-time information disclosure. Target: *Management Science*.

Presentations

Finite-time minimax bounds and an optimal Lyapunov policy in queueing control.

- 2026 INFORMS Annual Meeting, San Francisco, CA (Scheduled)
  Invited presentation, Machine Learning and Applied Probability Job Market Session — Nov 2026
- 16th POMS-HK International Conference, Shenzhen, China — Jan 2026
- 2025 INFORMS International Meeting, Singapore — Jul 2025

A non-convex optimization approach of searching algebraic degree phase-type representations for general phase-type distributions.

- 2025 INFORMS International Meeting, Singapore — Jul 2025

On substochastic inverse eigenvalue problems with the corresponding eigenvector constraints.

- 15th International Annual Conference of the Chinese Scholars Association for Management Science and Engineering (CSAMSE), Shenzhen, China — Jul 2023

Teaching and Seminar Experience

Stochastic Differential Equations Reading Seminar – Presenter Jul 2020 – Jul 2021

- Delivered a series of lectures based on Øksendal's *Stochastic Differential Equations: An Introduction with Applications*.

Markov Processes Reading Seminar – Primary Presenter Sep 2019 – Jan 2020

- Delivered most of the lectures based on Ethier and Kurtz's *Markov Processes: Characterization and Convergence*.

Honors and Awards

- Outstanding Graduate of Beijing (Top 5%) — 2023
- Outstanding Graduate of the Chinese Academy of Sciences (Top 5%) — 2023
- Loo-Keng Hua Scholarship, Chinese Academy of Sciences (Top 5%) — 2021